Stellar Cyber 6.6.0s is now live! Discover all the updates in the release notes!

Types of Stellar Cyber Sensors

Learn more at Stellar Cyber Academy.

The following link takes you to a course on the Stellar Cyber Academy technical training portal where you can learn more about this topic by watching the suggested lesson.

J01 Open XDR: Explained (01h:05m)

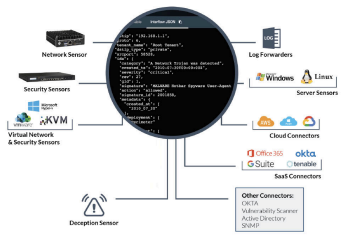
Modular Sensor Functions (12m:00s)

Learn about the role of modular sensors in gathering and analyzing data from diverse network sources. Discover how log forwarding, network traffic analysis, and intrusion detection contribute to real-time threat monitoring.

The first time you access a link on the portal during a session, you must log in to access content.

This topic describes the family of sensors available from Stellar Cyber, from lightweight server sensors that install as agents on target Linux and Windows servers to purpose-built physical sensors capable of monitoring at 10G network speeds. Sensors provide the data gathering foundation for Stellar Cyber's OpenXDR platform, gathering the right data with context.

Sensors are used to create visibility where it does not yet exist, or to simplify and consolidate telemetry collection with Stellar Cyber. You can capture data from the network and assets, all under the same license, and administered from the same platform.



Starting with the 4.3.7 release, new physical sensors are shipped as modular sensors on Ubuntu 22.04.

Family of Stellar Cyber Sensors

The Stellar Cyberfamily of sens consists of the following members:

- **Server Sensors** – Server Sensors install as software on target hosts to be monitored. They are also referred to as agents. Server Sensors are available for both Linux and Windows, as described below.
- **Device Sensors** – Device Sensor is an umbrella term that refers to any purpose-built Stellar Cyber sensor that includes both the host and the Stellar Cyber monitoring software. Device Sensors are provided as both physical devices (Photon sensors) and virtual machine images for different target environments.

> Previous releases provided a variety of different types of device sensors, including Network, Security, and Modular. Starting with 4.3.7, the only type of device sensor is Modular. You can use the Modular Sensor Profile to enable whatever sensor features you like, creating the same functionality provided by the different sensor types in previous releases. For your convenience, both Modular sensors and the legacy Network and Security sensors are described below.

Linux Server Sensor

A Linux server sensor is a managed background daemon that works as a modular sensor without log forwarding that also monitors:

- Process info
- Command execution
- Files
- File events

The server sensor converts that information to metadata and forwards it to the DP as Interflow. The DP can then correlate traffic, processes, users, and commands for security, DDoS, and breach attempt detections.

The server sensor launches the following processes:

- `aella_audit` – collects audit logs and provides file integrity monitoring
- `aella_conf` – handles the configuration
- `aella_ctrl` – monitors other services, and can stop or start them based on the configuration
- `aella_flow` – collects metadata in traffic
- `aella_mon` – collects system resource usage, including CPU, RAM, and disk

Data collected by the Linux server sensor can feed the following Stellar Cyber indices:

- Traffic (`aella-adr-*`)
- Linux Events (`aella-audit-*`)
- Sensor Monitoring (`aella-ade-*`)

Windows Server Sensor

The Windows Server Sensor (agent) runs as a Windows service in a compatible Windows Server system. The Server Sensor observes events within the Windows Server system and sends

Interflow data records to the data processor.

The captured events are:

- Hardware
- Security
- System
- Windows Firewall
- Windows Defender
- PowerShell

The Windows Server Sensor launches the following processes:

Name in Task Manager	Name in Services App	Description
aella_conf_win_srv.exe	Windows Agent Sensor Conf	Handles Windows Server Sensor configuration
aella_ctrl_win_srv.exe	Windows Agent Sensor Ctrl	Monitors other services, and can stop or start them based on the configuration
aella_winlog.exe	Windows Agent Sensor Logbeat	Sends Windows events to DP
aella_filebeat.exe	Windows Agent Sensor Filebeat	Sends log files from Windows DHCP server (if installed and running) to DP or acts as syslog forwarder if **Filebeat** is enabled in the Sensor Profile
aella_fim.exe	Windows Agent Sensor FIM	Provides file integrity monitoring services, as configured in the Sensor Profile
aella_diagnostics_win_srv.exe	Windows Agent Sensor Phonehome	Sends Windows Server Sensor logs to DP

Modular Sensor

See the top of this page for recommended lessons about this section at Stellar Cyber Academy.

A modular sensor lets you easily add the features you like to your sensor. This helps simplify your deployment and lets you manage the VM requirements for the sensors based on the modular features they use.

Modular Sensors always include log ingestion. From there, you can enable different features as part of your modular sensor profile:

- Enable the **Network Traffic** feature to monitor the virtual environment, the physical environment if connected to the span port of a physical switch, or the LAN segment via a mirror port on a switch. The sensor monitors network and server response times and can identify applications.

  The sensor converts that information to metadata and forwards it to the DP as Interflow. The DP can then provide security, DDoS, and breach attempt detections.

- Enable the **Sandbox** and **IDS** features to improve your security posture:
    - Sandbox lets  detect malware in files and network traffic through Stellar Cyber's integrated cloud service and also provides anti-virus services.
    - IDS lets you detect intrusion attempts using both files and network traffic.



Keep in mind that VM resource requirements increase as you add more features to the Modular Sensor Profile. Refer to **Virtual Modular Sensor Provisioning and Performance** for details on the resources required to run different combinations of features in a Modular Sensor Profile, as well as how to use the **show module** and **show module request** CLI commands to compare provisioned resources against those required to run specific feature combinations. Stellar Cyber only enables a Modular Sensor Profile on a sensor if the host VM's resources can support it.

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